Which Reroute Will an Airline Accept? Learning Route Preferences from Flight-Plan Revisions

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Abstract*—Every revision of a flight plan records a choice that an airline actually made, namely the route it dropped and the route it filed instead. Eighteen months of European traffic contain 1.48 million such pairs, which makes the archive a large and continuously refreshed sample of revealed route preference. This label is stronger than the one available to earlier route-ranking work: ranking the filed plan against generated alternatives that the airline may never have seen risks teaching a model only what a filed plan looks like, rather than which route an operator prefers. This paper fits a ranking model on the 1 134 000 of them that fall in the training period, expressing the key performance indicators of each route (i.e., flight time, distance, planned fuel and en-route charges) relative to those of the other route in the pair, and identifies the route that the airline chose in 68.1 % of held-out decisions, fifteen percentage points above the best single-feature rule. The model is also informative about its own uncertainty: on its most confident fifth of the cases it is right 94.9 % of the time, which is what allows a proposal channel to operate at a precision that is fixed in advance. As an outlook, simulating such a channel on the held-out quarter at the 80 % precision target surfaces 56 162 proposals, among which the airlines' own later filings confirm 20.1 kilotonnes (kt) of planned fuel in that quarter, or 63.5 kt of CO_2, which annualises to of order 220 to 250 kt on a planned-fuel basis. Room for improvement remains, and the clearest gap is the flights that never revise their plans at all, on which the model is not evaluated here; even so, these figures indicate that airline route acceptance is learnable from operational records alone. Worked examples of both successes and failures are reported, together with what the figures do and do not establish.

***Index Terms*—air traffic flow management, airline preferences, learning to rank, revealed preference, fuel efficiency, flight planning**

I. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the operator subsequently re-filed. The new route shed that delay completely, at a cost of 39 minutes of flight time and 1 397 kg of extra planned fuel. Both routes are held in the network's records and so, implicitly, is the operator's judgement that the delay was worth more than the 1.4 tonnes of fuel that avoiding it consumed.

That judgement is the quantity this paper models. A flight plan is a negotiated object: an airline files, the network responds with regulations and delay, and the airline may file again. Each revision records the route replaced and the route that replaced it.

The operational interest is concrete, and the network-level context is reported annually by the Performance Review Commission [1]: airlines frequently revise onto routes that *cost* fuel in order to escape delay, while fuel-cheaper alternatives for other flights are not identified in time to be filed. The difficulty is not generating candidate routes; it is knowing which candidate is acceptable to this particular operator, on this day, with this rotation to protect, and knowing it inside the planning window rather than under tactical time pressure. Needless to say, proposing routes that airlines reject is worse than proposing nothing at all, because a channel that is ignored is a channel that is eventually switched off.

The contribution of this paper is threefold. First, it shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it argues that the label construction is itself a design decision worth defending: a revision pair records a preference the airline actually expressed, whereas a generated candidate may never have been evaluated by the operator at all. Thirdly, it reports what the model has actually learned rather than only how well it scores, namely that route structure outweighs any single cost indicator, together with the caveats that such an attribution demands.

The remainder of this manuscript is organised as follows. Section II explains why a pair of routes drawn from a revision constitutes a preference observation and what a model of such observations is for, and Section III positions the work within the literature on rerouting and route choice. Section IV describes the archive, the split protocol and the model, while Section V reports the results. Finally, Section VI concludes and outlines future work.

II. Background

This section first describes fundamental concepts that are not a contribution of this work, but that are needed to follow the sections that come after it, namely how a flight-plan revision becomes a labelled preference pair and why that label is preferred to one built from generated candidates, and then states what the model is for. A reader already familiar with the filing process may move on to Section II-B.

A. *Flight-plan revisions as preference data*

In the European network, a filed flight plan may be revised before departure. When the revision changes the horizontal route, the system records a route change event containing the message that was replaced and the message that replaced it. Each event therefore yields two routes for the same flight, the same day and the same operator, exactly one of which the airline chose.

Treating this as a labelled preference pair requires care, because the two routes are separated in time as well as in geography: the replacement is by construction the later message, and any feature that drifts with time can therefore identify the answer without saying anything at all about the quality of the route. For instance, the time remaining to departure is necessarily smaller for the route that replaced the other, so a model given that feature in its raw form could answer "the one filed later" and be right on essentially every pair while knowing nothing whatever about routes. This is leakage rather than learning, and it would be invisible in the accuracy figure. To ensure a fair comparison between the two routes, all flight-level attributes were harmonised across the pair so that only route-level quantities differ, and the filing-time feature was collapsed to the pair minimum, so that both routes carry the smaller of the two values and the shortcut disappears.

When capacity is constrained, the network applies regulations to the affected volumes and assigns delay to the flights crossing them. An operator facing such a delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel and en-route charges. Neither answer is universally correct. For instance, a flight with a tight aircraft rotation may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the trade depends on operator-specific and flight-specific costs that the network does not observe, and that airlines weigh against published reference values for the cost of delay [2].

The archive reflects this trade directly. Slightly under a third of the revision pairs involve any attributed delay at all, and the newly filed route is more often longer, slower and more fuel-intensive than the one it replaced. Consequently, a proposal channel is not looking for the common case, but rather for the minority of flights on which a genuinely cheaper alternative exists and would be accepted.

The same events also settle how the training labels are built. Which of two routes an operator prefers cannot be asked at network scale, but it can be observed, and the labels used herein follow the logic of revealed preference (i.e., inferring what an actor values from the choice it made when a concrete alternative was available) [3]. The alternative construction, adopted by the authors' own earlier work, ranks the filed plan above generated candidates that the operator may never have seen or evaluated [4].

Ranking the filed plan above such candidates may therefore require only that the model recognise which route looks like a filed plan, which is a different and much easier skill than knowing which route an operator would prefer, and accuracy alone cannot distinguish the two. For instance, a generated candidate may cross airspace that the operator never uses, or follow a track that its own flight-planning system would discard on company policy; telling such a route from a filed plan calls for no knowledge of preference whatsoever. No dataset offers both label sources for the same flights, so the argument is about how the labels are built, not a measured comparison.

Revisions remove that difficulty (Fig. 1). Both routes were filed by the same operator, for the same flight, on the same day, so the negative is not a plausible alternative constructed after the fact but the route that the airline had committed to and then abandoned. The question also narrows: the model is not asked what an airline will file, but which of two specific routes it prefers, which is a strictly easier question and precisely the one a proposal channel needs answered.

The price to be paid is that revisions are not a random sample of flights, but rather flights that changed, which introduces a bias of its own. Rather than assuming that bias away, Section V-B measures it.

B. *Two operational use cases*

A model of route acceptance is worth building only if some decision changes when it exists, and two such decisions are stated here before any method is described. The first is a proposal channel: propose to an airline a route that is environmentally cheaper than the one it filed and that the model deems likely to be accepted, on the grounds that the operator may not have seen or evaluated that alternative in time. This is the use case that this paper simulates, by replaying held-out revisions and checking every proposal against the airline's own later filing (Sections IV-B and V).

The second use case is overload release. When a traffic volume is overloaded and some flights must be rerouted out of it, the flights to move first are those for which an acceptable and environmentally cheaper alternative exists, so that a single action relieves the overload, reduces environmental impact and limits the disruption imposed on the operators concerned. Network-side rerouting formulations acknowledge that they lack exactly this input, namely the likelihood that the operator would actually fly the assigned alternative [5], [6], [7], and the model presented herein supplies it. One could argue, however, that selecting flights for release by predicted acceptance could systematically burden particular operators, and any deployment would therefore need to monitor exactly that.

Both use cases return data to the model that drives them, in that every proposal answered, and every release accepted, is a new labelled observation, so the preference model improves with use. It should be stated plainly that the experiments reported herein test neither use case end to end: this paper demonstrates the preference model, simulates the first use case against later filings, and presents the second as something the model enables rather than demonstrates. Responses to proposals would, moreover, be a biased sample of flights, and a shadow-mode trial (i.e., one in which proposals are computed and logged but not sent) is the honest first step, a point taken up again where deployment is discussed.

III. Related work

Work on rerouting in air traffic flow management has largely approached the problem from the network side. Over two decades ago, the authors of [5] formulated rerouting as a dynamic network flow problem, and the authors of [6] subsequently added a stochastic model that reroutes flights as capacity evolves. Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled.

Almost concurrently, route choice in Europe was studied as a trade between fuel and en-route charges [8], which establishes that the choice is economic and therefore learnable, but which uses no label in which the operator rejected an alternative it had itself filed. The closest work to the present one is that of the authors of [7], who incorporate airline preferences into collaborative rerouting decisions and show that accounting for them changes which reroutes are selected. Those preferences, however, are inputs to an optimisation, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated. Similarly, the authors of [9] examine the value of airline collaboration in flow management, which motivates the proposal channel simulated herein but does not address which proposals would be taken up. Beyond the research literature, airlines already express their priorities through the user-driven prioritisation process [10], a legacy and trusted mechanism that a preference model of this kind would complement rather than replace.

The direct predecessor of this work applies learning to rank to flight routes for a different purpose: predicting the flight plans that have not yet been filed, so that traffic demand can be estimated earlier than the filing horizon allows [4]. That work established the framing adopted herein, namely a gradient-boosted ranker that scores routes by their indicators in context, and it reported the very difficulty that motivates the present paper. Its training pairs take the filed plan as preferred and a set of generated proposals (up to eleven route options produced by the Network Manager) as less preferred, and on the pre-tactical set none of the latter need match the filed route at all.

The case for building training pairs from revisions rather than from such generated sets is a design decision of this paper, and it is argued in Section II-A, where Fig. 1 contrasts the two constructions.

IV. Methodology

Figure 2 shows the procedure end to end; the two subsections that follow cover the data and the model. Thus, a reader interested only in the outcome can move on to Section V.

A. Data and pair construction

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs. Each route is described by twenty-three features, which are summarised in Table I.

All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements; accordingly, operator identity enters the model as a categorical feature only, and no operator, flight or airspace volume is identified anywhere in this paper.

Two feature-construction decisions deserve mention because they shaped the result more than the choice of learner. First, the four computed indicators enter as within-pair differences rather than magnitudes, obtained by subtracting the pair minimum from each, because absolute quantities are close to meaningless across the network: a 2 000 kg fuel figure means one thing on a short-haul leg and quite another on a long-haul sector, whereas the 300 kg by which one route exceeds the other carries the same meaning whether the sector is 400 or 4 000 km. Secondly, the ordered waypoint sequence is carried as text and encoded as a bag of tokens (i.e., each waypoint identifier is treated as a word and the route as the unordered collection of its words), which lets the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, and so be indistinguishable on every indicator, while only one of them crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot. Neither decision is novel in isolation, and Section V-B shows that the first of them matters far more than the authors expected.

Every revision pair is, by construction, a flight that *did* change route, and that creates a trap. A model trained only on such events learns that change is what happens, and recommends it far too eagerly against traffic that mostly stays, a bias that Section V-B measures. Put another way, the archive answers one question only, namely: when an airline changed its route, which route did it move to? It cannot answer the complementary question, because a flight that kept its route generates no record of the alternative that it declined.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of them was caught by a regulation while the other was not. The regulated flight had both a reason to move and a demonstrated alternative, namely the route that its twin was flying that same

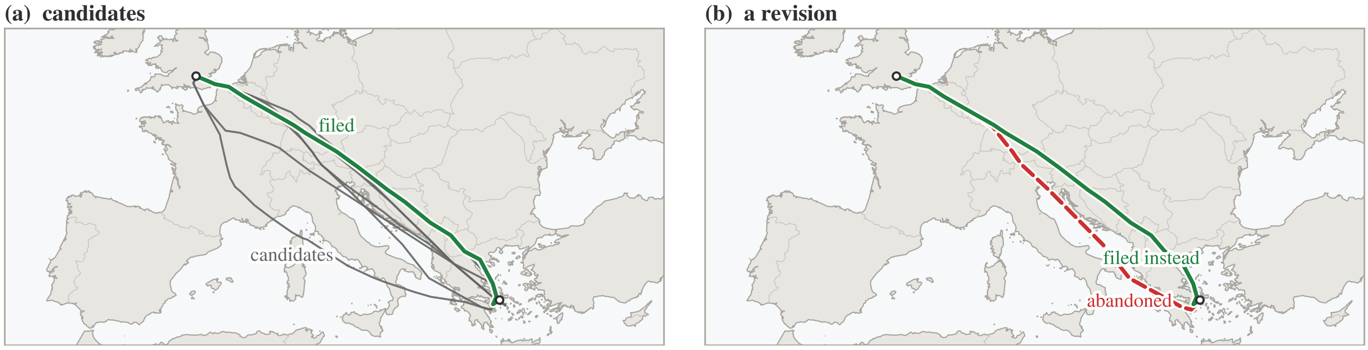

Fig. 1. Two ways to build a training pair, on one city pair. Left: the filed route (solid green) is ranked above candidates from a route generator (grey), which the airspace user may never have evaluated; those drawn here are other routes observed on the same city pair. Right: a revision supplies two routes that the same operator filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

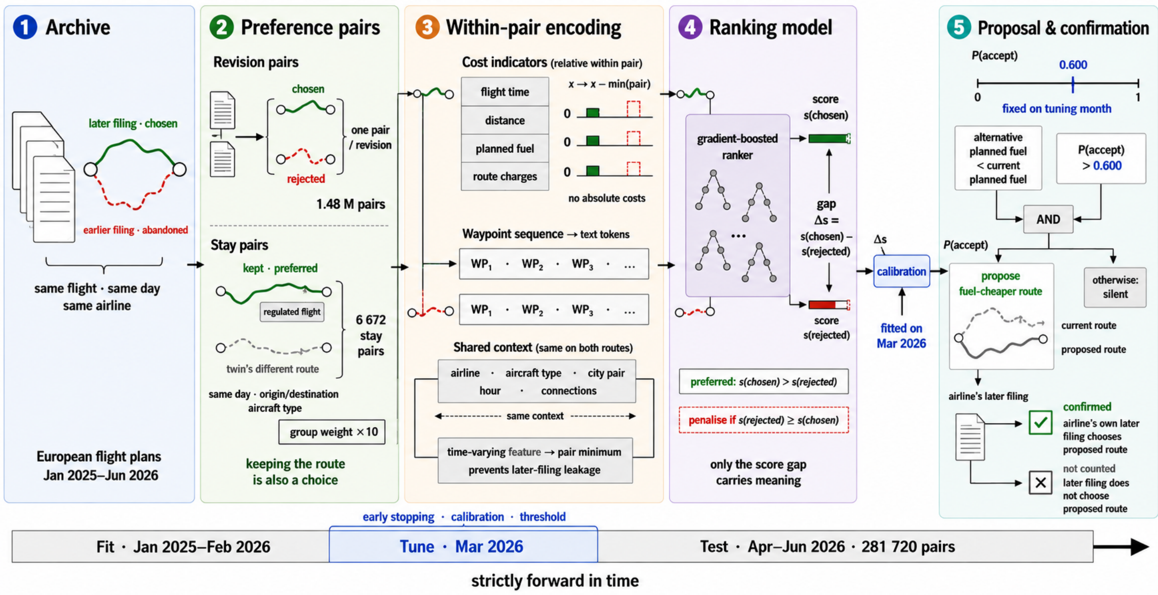

Fig. 2. How a filing archive becomes a proposal: the five stages of this section, followed for one flight, above the fit, tune and test periods.

day, and yet it stayed. Its route is accordingly labelled as preferred, and that of its twin as the alternative it declined. The confound arrives with the construction and belongs beside it: because the regulated flight is always the one whose route was kept, regulation status identifies the labelled answer. The conditions are restrictive and the yield is correspondingly low: 6 672 such *stay pairs*, of which 1 093 fall in the test months, against 1.48 million revision pairs. The two are used together, with stay pairs entering the fitting set at an increased group weight (the pair's weight in the ranking loss).

A rule as crude as "prefer the route of the flight that was regulated" is therefore right on every stay pair by construction, without examining a route at all, and a model with access to the delay features can score well on those pairs for the same empty reason. Accordingly, when the change-bias metric is reported, the seven regulation-derived features are masked at scoring time, which forces the decision onto route geometry and indicators. The masking happens at scoring time only: the figure says how well a model trained with the regulation features can decide once they are withheld, not what a model that never saw them would have learned instead. It should be noted that this is a repair applied after the fact rather than a clean design, and Section V-B quantifies how much it limits what can be concluded.

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026, tuned on March 2026, and evaluated once on April to June 2026 (281 720 pairs). Every threshold, calibrator and stopping decision is taken on the tuning month alone. Flights appearing in the tuning or evaluation months are removed from the fitting set.

B. Model, training and calibration

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Accordingly, the model gives each route a single number, which is not a probability and has no units, and which is only meaningful

TABLE I

WHAT THE MODEL SEES FOR EACH ROUTE; SOME ROWS SUMMARISE SEVERAL FEATURES (TWENTY-THREE IN ALL, SEVEN REGULATION-DERIVED). INDICATORS ENTER AS DIFFERENCES WITHIN THE PAIR; THE REST ARE IDENTICAL ON BOTH ROUTES AND ACT AS CONTEXT.

group	content
indicators	flight time, distance, planned fuel, route charges
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume
rotation	inbound and outbound connection times and states
context	operator, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction, in that for every historical pair the model is penalised whenever the abandoned route scores at least as high as the chosen one, and the penalty shrinks as the gap between them grows [11]. Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit.

Gradient-boosted decision trees are used as the scoring function [12], at a depth of 8, with the learning rate selected automatically and with early stopping on the tuning month. Stay pairs enter the fitting set with a group weight w, swept over $\{10, 30, 100\}$.

One detail proved consequential, and is worth recording here. When the fitting set is a weighted mixture of revision pairs and stay pairs, the tuning set must carry the same mixture at the same weights, since a tuning set composed only of revision pairs measures a different objective from the one being optimised. Unfortunately, this mismatch caused training to halt at the very first iteration and to return a below-chance model, which is a measurement failure that presents itself as a model failure. The pipeline therefore carries the same mixture into both splits, and warns when it cannot, so the models reported here are unaffected.

Once trained, the model has still to be connected to a decision. Roughly speaking, a bare score gap of, say, 3.7 in the model's own arbitrary units tells an operations room nothing at all, and before the model can be used to decide anything its scores have to be turned into statements of the form "about eight times in ten, an airline offered this route will take it". That conversion is what calibration provides, namely a monotone mapping from score gap to observed acceptance rate, fitted on the tuning month alone [13]. Three mappings were compared on the tuning month, namely the raw sigmoid of the score gap, Platt scaling and isotonic regression. Isotonic regression was selected on tuning-month log loss, and is what the pipeline uses. Miscalibration is common in this class of model [14], but it is worth reporting that on the held-out quarter the raw sigmoid was already well calibrated, at an expected calibration error (the mean gap between stated probability and observed rate) of 0.8 percentage points against 1.7 for isotonic regression, so that the fitted mapping is a

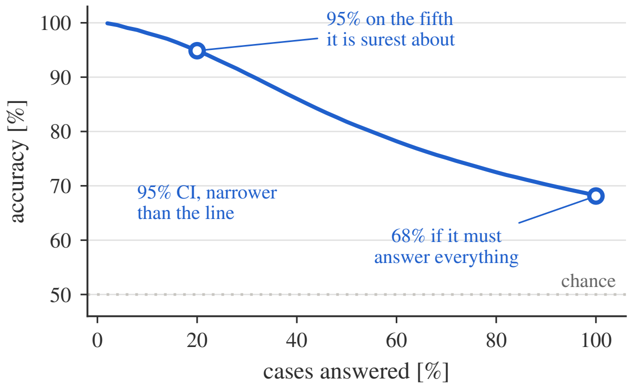

Fig. 3. Accuracy against coverage. Moving left, the model answers fewer cases and is right more often. A proposal channel is not obliged to answer every flight, and operates at the left of this curve.

safeguard here rather than a large correction. That outcome is reported rather than revised after the fact.

Calibration is also what makes a precision target meaningful, since without it one can rank proposals but cannot decide how many of them to make. On the held-out quarter, the stated and observed rates agree to within 1.7 percentage points on average. The proposal policy is then the following: for a flight whose candidate set (in this simulation, the other route of its revision pair) contains an alternative burning less planned fuel than the filed route, propose that alternative when its calibrated acceptance probability exceeds a threshold τ, with τ selected on the tuning month for a target precision and reported unchanged on the evaluation months. A proposal is counted as realised when the airline's own subsequent filing chose that alternative. For instance, at 80 % precision four proposals in five are confirmed in this way, and the fifth is simply not taken.

V. RESULTS

This section reports predictive performance, then the design choices and the change bias, then the errors and what the model weighs, and finally one proposal followed through to deployment.

A. *Predictive performance*

On the held-out quarter the model identifies the airline's chosen route in 68.1 % of 281 720 pairs (95 % interval 67.9–68.3, widening to 67.7–68.6 when flights on the same city pair are counted as one correlated group). Random choice scores 50 %, whereas the best single-feature rule, which prefers the route with less attributed delay, scores 53.2 %. Reassuringly, the tuning month scores 67.5 %, and there is therefore no sign of overfitting to it.

Accuracy alone understates operational value, because the model is informative about its own uncertainty. Fig. 3 reports accuracy when the model is permitted to abstain on those cases in which the two routes score close together. Let us call the share of flights that the model is allowed to answer its coverage, the remaining flights being left alone.

TABLE II

SCREENING ABLATIONS. EACH VARIANT IS RETRAINED FROM SCRATCH AT A REDUCED SETTING (DEPTH 6, 800 ROUNDS) THAT REACHES 60.6 % ON REVISION PAIRS; VALUES ARE THE CHANGE IN REVISION ACCURACY AGAINST THAT BASELINE, NOT HEADLINE ACCURACIES. A POSITIVE VALUE MEANS THE VARIANT SCORES *higher* ON REVISIONS THAN THE ADOPTED CONFIGURATION DOES. FOR THE STAY-PAIR CORRECTION THAT IS EXPECTED: IT IS THE PRICE PAID FOR THE CHANGE-BIAS REDUCTION REPORTED BELOW. DIFFERENCES ARE SIGNIFICANT UNDER MCNEMAR'S TEST, WHICH ASKS WHETHER TWO MODELS' DISAGREEMENTS ON THE SAME FLIGHTS FALL ASYMMETRICALLY RATHER THAN BY CHANCE, AT $p < 0.001$ EXCEPT THE WAYPOINT TEXT AT SCREENING SCALE ($p = 0.58$). THE MONOTONE CONSTRAINT (THE SCORE FORCED NEVER TO RISE AS THE NUMBER OF REGULATIONS ON A ROUTE GROWS) COSTS A FIFTH OF A POINT, WHICH IS WHY NONE IS IMPOSED.

variant	Δ accuracy [pp]
without within-pair KPI normalisation	-10.0
without stay-pair correction	$+3.7$
with monotone constraint	-0.2
without waypoint text	$+0.03$
including vertical-only revisions	-2.0

It is worth being explicit about what a margin of fifteen percentage points over a single-feature rule does and does not establish. It establishes that route choice carries structure beyond "prefer less delay", which was by no means self-evident. It does not establish, however, that 68.1 % is anywhere close to the ceiling of this task. A substantial share of the residual error falls on pairs in which the two routes are nearly indistinguishable on every recorded quantity, and another share on decisions driven by information that no route-level dataset contains (e.g., slot swaps, crew legality, curfews or company routing policy). Neither share, one case of which is inspected below, is separable from genuine model error without data that the network does not hold.

A model that scores well on average but decays from month to month is not deployable, and accuracy is therefore reported per month rather than pooled. Across the three test months it moves between 67.5 % and 69.0 %, and a replication run of the same recipe on 2025 alone scores 67.3 % and 67.5 %, at or just below the same band. Those six months span a schedule change, and accuracy does not step down across it. months rule out rapid decay, but they do not establish the retraining cadence that a deployed system would need.

B. Ablation and change bias

Table II reports screening ablations, each configuration removed in turn and the model retrained.

Expressing the indicators relative to the other route in the pair is decisively the most important choice, since removing it costs nearly ten percentage points, far more than any other component: it is what converts four absolute magnitudes into a comparison.

The waypoint-text result requires care, and illustrates why screening ablations can mislead. Interestingly, at screening scale the text feature appears worthless, and yet retrained at full scale keeping it is worth 1.4 percentage points (68.2 against 66.8), the reason being that a bag-of-words representation spread over thousands of rare tokens cannot repay its cost in 800 shallow rounds. It is a real but secondary contributor, worth about a tenth of the model's margin over the best single rule. Yet attribution ranks the same feature first, an apparent tension that Section V-C resolves.

The stay-pair correction whose price the table records buys the following. Measured on the held-out stay pairs, a model trained without correction prefers the alternative route in 65.2 % of cases where the flight kept its own, and when regulation features are masked so that only route understanding remains it scores 47.8 %, below chance. Adding stay pairs at group weight 10 raises the masked figure to 58.1 % (intervals $[55.1, 61.0]$ against $[44.8, 50.7]$, non-overlapping) at a cost of about one point on the revision metric.

Raising w from 10 to 100 moves three separate quantities. Stay accuracy with the regulation features visible rises from 90.3 % to 97.6 %; stay accuracy with them masked rises only from 58.5 % to 59.7 %; and revision accuracy falls from 68.2 % to 66.7 %. Most of what the additional weight buys is therefore the construction's giveaway (regulation status identifies the stayed route) rather than route judgement, and in this paper we use $w = 10$. On the masked stay metric, the sweep's own $w = 10$ run lands within half a point of the adopted model (58.5 against 58.1), which is the run-to-run variation these numbers carry, and that same variation separates the sweep's revision accuracy of 68.2 % from the 68.1 % headline.

C. Error analysis and feature attribution

Fortunately, errors concentrate where the model is already unsure: in 27.7 % of the errors the two routes scored within a quarter of the median margin, and only 5.6 % of the errors sit in the top quarter of the score margin, against 25 % if error were independent of confidence. This is the property that makes confidence filtering work.

The confident errors are nevertheless instructive, and one example from the held-out quarter is worth following. A business jet on a long-haul sector re-filed off a route carrying 124 minutes of attributed delay onto a route that carried no delay at all, saving 152 kg of fuel and about a minute of flight time; every indicator the model sees favoured the move, and yet the model confidently backed the abandoned route. The flight-level context is identical within a pair by construction, so the preference can only have come from the waypoint sequence itself, presumably a learned structural preference that overrode the indicators; confident errors are therefore not all of the hidden-information kind (decisions driven by facts, like a slot swap or a curfew, visible in no route-level record).

Opportunely, such cases cost less than their share of the error rate would suggest, because a proposal channel would have said nothing at all here: the alternative that it would have offered was the one already filed. The expensive errors are the opposite ones, in which the model confidently proposes a route that the operator declines, and the change-bias measurement reported above is the closest available proxy for their rate.

A ranking model that cannot be interrogated is of little use to an operational user who is asked to trust it, and Shapley attributions were therefore computed over the held-out

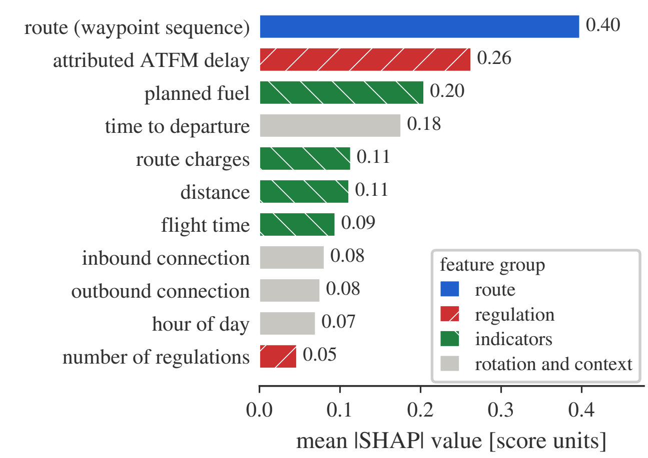

Fig. 4. Mean absolute Shapley value per feature on the held-out quarter. The waypoint sequence alone accounts for more of the score than the four cost indicators combined, and attributed ATFM delay outweighs planned fuel.

quarter. Principles from game theory can be used to interpret the prediction of a model for a given input, assuming that each feature is a player and that the model output is the payout. Let us consider the following scenario: all the features participate in the game (i.e., contribute to the model output) and enter the room where the game is played in a random order. The attribution of a feature is the average change in the payout received by the coalition already in the room when the corresponding player joins them, and this measure is the Shapley value [15]. Averaging their magnitude over the quarter gives Fig. 4, which is a ranking not of what correlates with acceptance, but of what moves the score of this particular model.

Two results stand out. First, the waypoint sequence is the single largest term, ahead of every cost indicator, which means that route structure carries information about the choice that the four cost indicators do not, and which is precisely what a cost-only model cannot represent and a learned one can. Secondly, attributed ATFM delay moves the score more than planned fuel does, which is consistent with delay being directly monetisable at planning time whereas the fuel difference between two candidate routes is often small beside it. This is a property of the score of the model rather than a statement about how any operator reasons.

Descending to individual waypoints sharpens the picture. Deleting a single identifier from a route and rescoring shifts the score by up to 2.5 points for the most influential one, and the affected waypoints cluster in a few places along the track rather than spreading evenly, which indicates that what the model has acquired is route-specific structure rather than a general preference for shorter routes. These are statistical regularities in filing behaviour across the network; they carry no assessment of any airspace, and individual waypoints and volumes are deliberately not reported.

One caveat should nevertheless be carried forward. The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy. Attribution measures how much a feature moves the score of the model we have; ablation measures how much worse a model without it would be. Where features are correlated, as route geometry and cost indicators plainly are, the two answer different questions and the second is the one that governs what may safely be dropped.

D. A worked proposal and the conditions for deployment

One case from the held-out quarter shows what the channel is for, and is worth following from end to end, because every step is a step that the deployed system would take.

A narrow-body was filed on a medium-haul sector, with no regulation anywhere in the picture. First, the system considers the filed route and one alternative. It then computes their indicators and subtracts the pair minimum from each, so that what reaches the model is a difference rather than an absolute quantity, and finds that the alternative burns 2 016 kg less planned fuel, about a fifth of the trip fuel, at essentially the same flight time. Both routes carry the same flight-level context, namely the same operator, the same aircraft, the same departure hour and the same inbound and outbound connections. Needless to say, nothing that the model sees distinguishes the two routes except where they go and what they cost.

The model then scores each route, and the alternative comes out $+3.7$ ahead on that same unitless scale. That margin is not yet a decision: it passes through the calibrator fitted on the tuning month, which converts it into an acceptance probability, and that probability is finally compared with the threshold $\tau = 0.600$ that was fixed before the evaluation period began. This case clears the threshold, and the policy therefore proposes. Had the margin been smaller, the flight would simply have been left alone: the system is permitted to say nothing, and on roughly four flights in five that is what it does.

The operator's own later filing chose exactly that route. Keep in mind, however, that the airline was told nothing, so this is evidence not that a proposal would have been accepted, but that the model identified, before departure, a route the airline itself went on to prefer. That is the strongest validation available without running the channel, and the form on which every fuel figure in this paper rests.

One proposal followed end to end says little about a channel making tens of thousands of them, and this is in any case a feasibility study, in which no operational service is proposed. Two conditions stand between the results above and an operational channel, and neither of them concerns the model itself. A third, that acceptance itself is unobserved, is taken up as future work below.

The first is the freshness of the regulation picture. Attributed delay and filing lead time are among the strongest inputs, and both are properties of a moment rather than of a flight. A score computed against a regulation state that has since changed is a score answering a question that nobody asked, and the binding constraint at serving time is therefore data currency rather than computation: the scoring step is a single pass through a fixed tree ensemble.

The second condition concerns the population itself, which changes. A deployed channel would see every flight, and most flights never revise their plan at all, so the model would be asked a question on which it was not evaluated. The honest expectation is that precision falls, because the flights that never revise are precisely the ones whose operators were content with what they filed, and Section V-B quantifies how much of that gap the stay pairs close: some of it, not enough.

Fortunately, a channel also generates exactly the data that would close this second gap, since every proposal answered is a new labelled observation of the very kind that the archive cannot supply (Section II-B); as those responses form a biased sample of the flights that the channel itself selected, the shadow-mode trial proposed as future work comes first.

What such a channel could be worth is reported here only as an outlook. At an operating point fixed in advance for 80 % precision ($\tau = 0.600$, delivering 79.2 % on the evaluation months), the simulated channel surfaces 56 162 proposals over the held-out quarter carrying 20.1 kt of planned fuel that the airlines' own later filings confirm, equivalent to 63.5 kt of CO_2 and, extrapolated, of order 220 to 250 kt of CO_2 per year. These figures are computed on flights that revised their plans, and planned fuel is not burned fuel, so they should be read as what a channel could have surfaced earlier rather than as a traffic-wide counterfactual. The full channel analysis, covering operating points, value concentration and sensitivity, is the subject of a journal extension of this work.

VI. Conclusions

Airline route acceptance is learnable from flight-plan revisions at network scale, and the findings of this paper are threefold. First, a model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases, fifteen percentage points above the best simple rule, and supports operation at a selected precision rather than at a single accuracy figure: on the most confident fifth of the decisions it is right 94.9 % of the time. Secondly, much of what the model can learn is decided before training, by the label construction: a revision pair records a preference that the airline actually expressed, whereas ranking a filed plan above generated candidates can be solved by recognising filed plans alone. Thirdly, the model weighs route structure more heavily than any individual cost indicator, which is precisely the information that a formulation based on costs alone cannot represent.

One practical note should be made before turning to the limits. The number that characterises this system is not 68.1 % but the curve of Fig. 3, because a channel chooses where on that curve to sit. Note that ranking a pair correctly and proposing correctly are different quantities, and that the second of them is the operational one: at the threshold used here, the channel proposes on about a fifth of the eligible flights, and 79.2 % of those proposals match what the airline went on to file. Accordingly, any first deployment would sit well to the left of that curve, and would move right only as acceptance is observed.

One of the deployment conditions set out above also bounds what this work can claim scientifically: the data on flights that stayed is constructed in such a way that regulation status identifies the answer, so what the model knows about keep-or-change decisions cannot be measured cleanly against the archive as it stands. Future work should therefore be directed, in the first place, at assembling a balanced stay dataset in which regulation status is independent of the label, which requires data that is not currently assembled rather than data that does not exist. Last but not least, a shadow-mode trial logging the real responses of operators would close the remaining gap.

Disclaimer

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